In [39]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score
```

In [40]:
```
df = pd.read_csv('/home/backup/ml_datasets/pima_indians_diabetes.csv')
df['class'] = df['class'].map({'tested_positive':1,'tested_negative':0})

X = df.drop('class', axis=1)
y = df['class']
print(df.head(5))
```

```
   preg   plas  pres  skin   insu  mass   pedi   age  class
0   6.0  148.0  72.0  35.0    0.0  33.6  0.627  50.0      1
1   1.0   85.0  66.0  29.0    0.0  26.6  0.351  31.0      0
2   8.0  183.0  64.0   0.0    0.0  23.3  0.672  32.0      1
3   1.0   89.0  66.0  23.0   94.0  28.1  0.167  21.0      0
4   0.0  137.0  40.0  35.0  168.0  43.1  2.288  33.0      1
```

In [42]:
```
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_state=4

print(f"Training set size: {X_train.shape[0]}")
print(f"Testing set size: {X_test.shape[0]}")
```

```
Training set size: 614
Testing set size: 154
```

In [43]:
```
model = LinearRegression()
model.fit(X_train, y_train)
print(f"Linear Regression \n")
#print(f"Intercept: {model.intercept_}")
#print(f"Coefficient: {model.coef_} \n")
print(y_test[:5])

y_pred = model.predict(X_test)
print(y_pred[:5])
```

```
Linear Regression

668    0
324    0
624    0
690    0
473    0
Name: class, dtype: int64
[0.33550028 0.23809869 0.1510522  0.2401365  0.48142376]
```

In [44]:
```
linear_mse = mean_squared_error(y_test, y_pred)
print(f"Mean Squared Error: {mse}")

linear_r2 = r2_score(y_test, y_pred)
print(f"R-squared score: {r2}")
```

```
Mean Squared Error: 0.171045272808501
R-squared score: 0.2550028117674177
```

In [45]:
```
ridge = Ridge()
ridge_parameters = {"alpha":[0.1,1,10]}
ridge_cv = GridSearchCV(ridge, ridge_parameters, cv=5)
ridge_cv.fit(X_train, y_train)
ridge_model = ridge_cv.best_estimator_
ridge_pred = ridge_model.predict(X_test)
ridge_mse = mean_squared_error(y_test, ridge_pred)
ridge_r2 = r2_score(y_test, ridge_pred)
```

In [46]:
```
lasso = Lasso()
lasso_parameters = {"alpha":[0.001,0.01,0.1]}
lasso_cv = GridSearchCV(lasso, lasso_parameters, cv=5)
lasso_cv.fit(X_train, y_train)
lasso_model = lasso_cv.best_estimator_
lasso_pred = lasso_model.predict(X_test)
lasso_mse = mean_squared_error(y_test, lasso_pred)
lasso_r2 = r2_score(y_test, lasso_pred)
```

In [65]:
```
print("Best Ridge Alpha: ", ridge_cv.best_params_["alpha"])
print("Best Lasso Alpha: ", lasso_cv.best_params_["alpha"])
print("\n")
print("                MODEL PERFORMANCE")

print("-"*48)
print(f"{'Model':<20} {'MSE':<15} {'R2 score'}")
print("-"*48)
print(f"{'Linear Regression':<20} {linear_mse:<15.2f} {linear_r2:.4f}")
print(f"{'Ridge Regression':<20} {ridge_mse:<15.2f} {ridge_r2:.4f}")
print(f"{'Lasso Regression':<20} {lasso_mse:<15.2f} {lasso_r2:.4f}")
print("-"*48)
```

```
Best Ridge Alpha:  10
Best Lasso Alpha:  0.001


                MODEL PERFORMANCE
------------------------------------------------
Model                MSE             R2 score
------------------------------------------------
Linear Regression    0.17            0.2550
Ridge Regression     0.17            0.2526
Lasso Regression     0.17            0.2533
------------------------------------------------
```

In [ ]: